An Integrated Vision and Sensor Framework for Swimming Pool Safety Monitoring and Usage-Aware Maintenance Scheduling

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***Abstract*—Swimming pool operators must simultaneously supervise bather safety, water chemistry and pool cleanliness, and these tasks are typically performed by separate manual procedures on fixed schedules that do not reflect actual pool usage. This paper presents an integrated monitoring framework that combines four independently trained models behind a single shared camera stream and one decision layer. A one-class detector estimates the number of swimmers per frame and converts the resulting occupancy into accumulated bather load in person-hours, which drives a transparent rule-based maintenance schedule. A second detector classifies swimmer behaviour and raises a drowning alert only after three consecutive positive frames. A third detector identifies foreign objects in the water while excluding legitimate pool items. A gradient-boosted classifier maps five water-chemistry readings to a three-level safety state. The behaviour detector achieved a mean average precision at an intersection over union threshold of 0.50 of 0.907, and the object detector achieved 0.903, on their respective held-out test sets. The system is presented as an operator decision-support prototype rather than an autonomous safety controller.**

***Keywords*—bather load, drowning detection, object detection, pool monitoring, predictive maintenance, water quality classification**

I. Introduction

Public and hotel swimming pools are operated under three simultaneous obligations: preventing drowning incidents, maintaining water chemistry within safe limits, and keeping the water free of foreign objects. In current practice each obligation is discharged by a separate mechanism — lifeguard observation, periodic manual or sensor-based water testing, and scheduled physical cleaning. These mechanisms operate independently, produce records in incompatible formats, and give the operator no single view of the pool's overall condition [CITATION NEEDED].

Two limitations follow from this separation. First, safety supervision depends on sustained human attention, which degrades over long shifts and across large water surfaces. Second, maintenance is commonly scheduled by calendar rather than by usage, so a pool that has been heavily used receives the same treatment interval as one that has been nearly empty. The consequence is either premature treatment, which wastes chemicals and staff time, or delayed treatment, which allows water quality to degrade before the next scheduled intervention [CITATION NEEDED].

Deep object detection has been applied to aquatic environments for swimmer counting and for drowning-posture recognition [CITATION NEEDED]. However, the reported systems generally terminate at the perception stage: they output a count, a class label, or an alert. The step from a perception output to an operational decision — specifically, from an occupancy estimate to a maintenance action — is largely unaddressed. Similarly, water-quality classification has been studied as a standalone tabular learning problem [CITATION NEEDED], without coupling to the visual evidence of how heavily the pool was actually used during the interval being assessed.

The gap this paper addresses is therefore integration and actionability: no reported system places swimmer counting, behaviour recognition, foreign-object detection and water-chemistry classification behind one shared acquisition layer and one decision layer, and none converts the perception output into an explainable maintenance recommendation.

The contributions of this paper are as follows.

1) Integrated architecture: three vision models consume a single shared camera stream under a staggered inference

schedule, and a fourth sensor-driven module runs in parallel, with all outputs written into one thread-safe system state (Section III-A).

2) Usage-aware maintenance formulation: per-frame swimmer counts are converted into accumulated bather load measured in person-hours and mapped onto a transparent threshold-based maintenance schedule with three priority levels (Section III-B).

3) Cross-module decision layer: occupancy, behaviour, cleanliness and water-chemistry states are combined into a single overall risk level and maintenance urgency (Section III-F).

4) Empirical comparison of training data: single-source and combined-source swimmer datasets are compared to assess the effect of source heterogeneity on detection performance (Section V-A).

5) Honest evaluation and limitations analysis: test-set sizes, the reproducibility status of each checkpoint, and the conditions under which each module should not be relied upon are reported explicitly (Sections V–VII).

The evaluation is organised around three research questions. RQ1: can a single-class swimmer detector produce occupancy estimates stable enough to drive a usage-based maintenance schedule? RQ2: does combining swimmer datasets from heterogeneous sources improve detection generalisation relative to a single-source dataset? RQ3: can four independently trained models be executed on one shared frame stream within the latency budget of an interactive operator dashboard?

Section II reviews related work. Section III presents the architecture and each module. Section IV describes the experimental setup. Section V reports results against RQ1–RQ3. Section VI discusses the findings and Section VII the limitations. Section VIII concludes.

II. Related Work

A. Swimmer Detection and Crowd Estimation in Aquatic Scenes

Water introduces failure modes absent from general pedestrian detection: partial submersion, specular reflection, surface distortion, and severe scale variation between near and far swimmers [CITATION NEEDED].

[CITATION NEEDED] Three to four works. For each, state the detector family, the dataset and the reported metric, then state what it does not do, so the gap accumulates.

B. Drowning and Anomalous Behaviour Recognition

[CITATION NEEDED] Three to four works, including at least one that discusses the precision–recall trade-off in safety-critical alerting, since Section VI argues for recall priority.

C. Foreign-Object and Debris Detection in Pools

[CITATION NEEDED] Two to three works. If pool-specific literature is thin, cite the closest underwater or floating-debris detection work and state explicitly that pool-specific work is limited.

D. Machine Learning for Water-Quality Classification

Gradient-boosted decision trees are a standard choice for tabular classification over sensor readings, and XGBoost [2] is widely used for this class of problem.

[CITATION NEEDED] Two to three works applying learned classifiers to pool or recreational water chemistry, as distinct from drinking-water potability.

E. Usage-Based and Predictive Maintenance

[CITATION NEEDED] Two to three works on condition-based or usage-based maintenance scheduling in facilities management, to position the bather-load contribution.

F. Positioning

TABLE I
Comparison of Related Systems

System	Mod.	Shared	Behav.	Water	Maint.
[Ref]	—	—	—	—	—
[Ref]	—	—	—	—	—
[Ref]	—	—	—	—	—
This work	4	Yes	Yes	Yes	Yes

Populate one row per cited system once Section II citations are chosen.

To the best of the authors' knowledge, no prior system in Table I combines all four capabilities behind a single acquisition and decision layer.

III. System Design

A. Architecture Overview

[FIGURE REQUIRED] Fig. 1 must be drawn before submission: architecture of the proposed framework, from shared frame acquisition to the operator dashboard. Caption below the figure, sentence case, ending in a full stop, using the abbreviation "Fig." Export as vector or at least 300 dpi. Do not screenshot the dashboard.

The framework is organised as four independent inference modules, one shared acquisition layer, one shared state object and one decision layer. A single camera source — a universal serial bus webcam, a network camera or a recorded video file — is opened once by the acquisition service. Each acquired frame is dispatched to the vision modules according to a staggered schedule, so that no single frame incurs the cost of three forward passes. The water-quality module runs on an independent timer driven by sensor readings rather than by frames.

Each module returns a structured result dictionary and writes it into its own section of a thread-safe system state. No module reads or writes another module's section. The decision layer

reads the complete state snapshot and derives two facility-level outputs: an overall risk level and a maintenance urgency level. A server based on Flask and Socket.IO emits the annotated frame and the state snapshot to a browser dashboard.

This design has two properties worth stating. First, the modules are independently replaceable: each was trained separately and each can be evaluated in isolation without the dashboard. Second, the coupling between modules is confined to a single component, which makes the cross-module logic auditable.

B. Component 1: Swimmer Detection and Usage-Aware Maintenance

1) Detection: A single-class object detector of the YOLOv8 family [3] is fine-tuned to detect the class swimmer. At inference the detector is applied with a confidence threshold and a non-maximum suppression intersection over union threshold; overlapping boxes for the same swimmer are suppressed by non-maximum suppression.

2) Region-of-interest filtering: An optional polygon expressed in coordinates normalised to the interval [0, 1] delimits the water surface, so that the same polygon scales with frame resolution. A detection is accepted only if the bottom-centre point of its bounding box falls inside or on the polygon. The bottom-centre is used rather than the box centre because it approximates the point at which a person contacts the water or the deck, which is the discriminating cue for people standing at the pool edge.

3) Count stabilisation: Raw per-frame counts fluctuate when a detection sits near the confidence threshold or when a swimmer is briefly occluded. The system therefore maintains a sliding window of the five most recent raw counts and reports their median. This suppresses single-frame excursions at the cost of a bounded delay before a genuine occupancy change is reflected.

4) Occupancy and density: Maximum capacity is derived from the pool surface area and the area required per bather,

$$C = \lfloor A / a \rfloor \tag{1}$$

where A is the pool surface area in square metres and a is the surface area required per bather in square metres. The instantaneous occupancy ratio is

$$O = (n / C) \times 100 \tag{2}$$

where n is the stabilised swimmer count. Density is reported on four levels: low below 50%, moderate at or above 50%, high at or above 80%, and over capacity at or above 100%.

[VERIFY] The implementation uses defaults of A = 250 m² and a = 2.7 m², giving C = 92. These are configurable defaults, not facility-approved figures. Before submission, either cite the standard from which a = 2.7 m² is taken, or state in the text that the value is a configurable parameter and that no standard is claimed. Do not cite a standard you have not read.

5) Bather load: For each reading interval, occupancy is integrated over time. For a single reading,

$$s_i = n_i \, \Delta t_i \tag{3}$$

where s is person-seconds, n is the stabilised count and Δt is the elapsed interval in seconds. Accumulated bather load in person-hours over a session is

$$L = (1 / 3600) \, \Sigma \, s_i \tag{4}$$

Bather load is a measure of occupancy integrated over time, not a count of unique individuals; ten swimmers present for half an hour and five swimmers present for one hour both yield five person-hours.

6) Maintenance scheduling: Accumulated load is compared against per-action thresholds. The implemented thresholds are 20 person-hours for chlorine dosing, 30 for skimmer cleaning, 50 for filter backwash, 100 for shock treatment and 200 for deep cleaning. Priority escalates with the ratio of accumulated load since the last completion to the threshold: medium at 1.0, high at 1.5 and critical at 2.0. Completing an action resets that action's reference load to the current load.

The scheduler is deliberately rule-based rather than learned. This makes every recommendation traceable to the specific threshold that produced it, which is a prerequisite for operator trust and for post-hoc audit.

[VERIFY] The five thresholds are implemented project rules. They have not been validated against a facility maintenance policy or operational log. This must be stated explicitly and must not be presented as established practice unless a source can be cited.

C. Component 2: Water-Quality Classification

Five chemistry variables — pH, temperature, free chlorine, turbidity and total dissolved solids — are read from a microcontroller over a serial link, with a simulated fallback when hardware is unavailable. Readings are scaled and passed to a gradient-boosted tree classifier trained with XGBoost [2], which assigns one of three states: safe, warning or critical. Label encoding, feature scaling and the trained model are persisted as separate artefacts and loaded at runtime.

D. Component 3: Swimmer Behaviour and Drowning Detection

A three-class detector of the YOLOv8 family distinguishes swimming, drowning and person out of water. Because a per-frame drowning classification is noisy, an alert is raised only after the drowning class is detected in three consecutive processed frames. This rule trades detection latency against alert stability: it suppresses isolated false positives at the cost of delaying a genuine alert by the duration of two additional processed frames.

[VERIFY] State the delay introduced by the three-frame rule in seconds. It depends on the processed-frame rate, which is itself unmeasured (Section IV). A safety-oriented reviewer will ask for this number.

E. Component 4: Foreign-Object Detection

A six-class detector based on YOLOv8n identifies objects on and in the water. Application logic partitions the classes into true garbage — aluminium foil, bottle, juice container and thermocol — and legitimate pool items, specifically ball and leaf, which are detected and displayed but never raise an alert. This class disambiguation is a deliberate false-positive mitigation: a toy ball in a pool is an expected object, and alerting on it would train operators to ignore the alert channel.

F. Cross-Module Decision Layer

The decision layer is the only component that reads more than one module's state. Overall risk is raised by an active drowning alert, and further raised when occupancy is high or over capacity, on the reasoning that a crowded pool both increases incident likelihood and reduces the visibility of any individual swimmer. Maintenance urgency is raised when pending crowd-driven maintenance actions coincide with a water-quality warning or critical state, on the reasoning that accumulated bather load and degraded chemistry are corroborating rather than independent signals. This is the component that distinguishes the system from four detectors sharing a screen.

G. Contribution-to-Evaluation Traceability

TABLE II
MAPPING FROM CONTRIBUTIONS TO EVALUATION

Contribution	Designed	Evaluated
Integrated architecture	III-A	V-E (RQ3)
Usage-aware maintenance	III-B	V-A (RQ1)
Cross-module decision layer	III-F	V-E
Single vs. combined data	III-B	V-A (RQ2)
Limitations analysis	—	V–VII

IV. EXPERIMENTAL SETUP

[VERIFY — SECTION BLOCKING] This section cannot currently be written truthfully. The four models were trained on four different machines with no common protocol, and the required records were not recoverable from the project repository. Every item below must be recovered from the training machines before submission. If a value cannot be recovered, the correct action is to re-run the experiment, not to estimate the number.

A. Hardware and Software Environment

The integrated system runs a Python backend using Flask and Socket.IO and is served to a browser client.

[VERIFY] Required per module, stated separately: training device (CPU/GPU and memory), training duration, Python version, framework and version (Ultralytics for Components 1, 3 and 4; XGBoost for Component 2), and random seed. Also required: the deployment machine specification and the server port, which is documented inconsistently in the project as 5000 and 5051.

B. Datasets

TABLE III
DATASETS USED

Module	Source	Licence	Cls.	Test
C1 v1	[VERIFY]	[VERIFY]	1	[VERIFY]
C1 v2	[VERIFY]	[VERIFY]	1	[VERIFY]
C2 water	Synthetic	[VERIFY]	3	[VERIFY]
C3 drown	[VERIFY]	[VERIFY]	3	[VERIFY]
C4 garbage	Roboflow [4]	CC BY 4.0	6	25

Train and validation split sizes are also required for every row and are omitted here only for column width.

All detection datasets follow the standard YOLO layout, with one label file per image and bounding boxes normalised to the interval [0, 1]. The Component 4 dataset is distributed under CC BY 4.0, which makes attribution a licence condition rather than a courtesy; reference [4] must be cited at first mention of that dataset.

[VERIFY — HIGHEST PRIORITY] Component 1 dataset provenance is the single most important missing item in this paper. An internal project note refers to a "single-source KSU-style dataset" for v1; that phrase is not a citation and must not appear in the paper. Required for both v1 and v2: dataset name, owner, retrieval URL, licence, images per split and number of labelled swimmer instances. Without these, Table III cannot be completed and Section V-A cannot be defended.

C. Training Configuration

The swimmer detector was trained for 50 epochs at an input resolution of 640 pixels, with a batch size of 8, an early-stopping patience of 15 epochs and 2 data-loader workers. The drowning detector was trained for 100 epochs. The object detector was trained for 10 epochs on a central processing unit.

Reproducibility statement. The two deployed swimmer checkpoints carry embedded metadata identifying a YOLOv8n starting checkpoint, whereas the training script in the repository initialises from YOLOv8m and exports under different filenames. The script as committed therefore does not reproduce the deployed checkpoints. This is disclosed rather than concealed, and reconciling the script with the deployed checkpoints is listed as immediate future work. Additionally, the source files implementing bather-load accumulation and maintenance scheduling were reconstructed from compiled bytecode after the original sources were lost; class names, method signatures, variables and thresholds are preserved. This is disclosed for completeness.

D. Evaluation Metrics

Detection performance is reported as precision, recall, mean average precision at an intersection over union threshold of 0.50 (mAP@50) and mean average precision averaged over thresholds from 0.50 to 0.95 (mAP@50–95). Precision is the proportion of predicted boxes that are correct; recall is the

proportion of ground-truth objects that are detected; mAP@50–95 is the stricter measure and penalises imprecise box placement. Classification performance for the water-quality module is reported as accuracy and macro F1-score.

[VERIFY] Two metric families are missing. First, an occupancy-level metric: detection metrics measure box quality, not count quality, while RQ1 is a counting question. Report the mean absolute error between the stabilised count and a manually annotated ground-truth count over a held-out video segment. Second, runtime metrics: frames per second, per-module inference time and end-to-end latency on the deployment machine. RQ3 cannot be answered without them.

V. Results

A. Component 1: Swimmer Detection (RQ1, RQ2)

TABLE IV
Swimmer Detection Performance on the Held-Out Test Set

Model	Data	P	R	mAP50	mAP50-95
v1	Single	0.777	0.760	0.794	0.338
v2	Combined	0.785	0.765	0.793	0.334

Test-set size: [VERIFY]. These values are unconfirmed — see the verification note below.

[VERIFY — BLOCKING] These values are currently unsafe to publish. They are the figures carried in the project presentation, but no raw evaluation output maps them to a named checkpoint, and a separate internal note records entirely different figures (approximately 85.5% and 80.6%) for the same two models. Two mutually inconsistent accounts of the same experiment exist and neither is supported by a saved results file. Before submission, run the evaluation separately for each checkpoint on the same test split, at the same input resolution and confidence threshold, save the raw output, and repopulate this table from it. The existing evaluation script selects the first checkpoint it finds and must be modified to take the checkpoint path as an argument.

Interpretation, valid for the values shown. The two models are close. The combined-source model attains marginally higher precision and recall; the single-source model attains marginally higher mean average precision at both thresholds. The difference at mAP@50 amounts to 0.001, which is within the range attributable to run-to-run variation and cannot support a claim that either model is superior.

Answering RQ2 honestly, the result does not demonstrate that combining sources degrades performance, and it does not demonstrate that combining sources helps. What it demonstrates is that adding a second data source produced no measurable improvement, which is itself a reportable negative result. Plausible explanations include differences in camera geometry, illumination, pool appearance, swimmer scale and annotation convention between the two sources. Establishing which of these dominates would require a controlled ablation over source subsets, which was not performed.

[VERIFY] RQ1 is not yet answered. Detection metrics alone do not establish count stability. Add the count mean-absolute-error experiment described in Section IV-D.

B. Component 3: Drowning Detection

TABLE V
Behaviour Detection Performance

Metric	Value
Precision	0.877
Recall	0.883
mAP@50	0.907
mAP@50–95	0.594
Epochs	100

Test-set size: [VERIFY].

This is the strongest-performing module. A recall of 0.883 means that approximately 12% of ground-truth drowning instances are missed on a per-frame basis. Because the alert rule requires three consecutive positive frames, the effective miss rate for a sustained drowning event is lower than the per-frame figure, but this has not been measured directly.

[VERIFY] Measure the event-level miss rate, not only frame-level recall. For a safety module this is the number that matters.

C. Component 4: Foreign-Object Detection

TABLE VI
Object Detection Performance

Metric	Value
Precision	0.853
Recall	0.878
mAP@50	0.903
mAP@50–95	0.393
Epochs	10 (CPU)
Test images	25

Two observations follow. First, the gap between mAP@50 (0.903) and mAP@50–95 (0.393) indicates that the model localises objects well enough to be counted correct at a loose overlap criterion but places boxes imprecisely. Second, and more consequentially, a test set of 25 images is too small to support a confident performance claim; the confidence interval around each figure is wide.

D. Component 2: Water-Quality Classification

The classifier attains an accuracy of 1.00 and a macro F1-score of 1.00 on the augmented dataset. This result is presented with its caveat in the same paragraph rather than deferred. A perfect score on a synthetically augmented dataset indicates that the

three classes are separable by simple thresholds on the input variables, which is the expected outcome when the labels were themselves generated from threshold rules. The result therefore validates the implementation pipeline rather than demonstrating generalisation.

[VERIFY] Three experiments are required before this module can be claimed as a contribution: a comparison against an explicit threshold baseline, to establish whether the learned model adds anything over the rules that generated the labels; evaluation on noisy or out-of-range inputs; and k-fold cross-validation. Without the first in particular, a reviewer is entitled to conclude that the classifier reimplements an if-statement, and this is the most probable single point of attack on the paper.

E. Integrated System (RQ3)

[VERIFY — BLOCKING] This subsection cannot currently be written. RQ3 concerns whether four models can run on one shared stream within an interactive latency budget, and no timing data exists. Report end-to-end latency, achieved frame rate with and without staggered scheduling, per-module inference time and memory footprint on the deployment machine. Until these are measured, either report them or remove RQ3 from Section I — an unanswered research question stated in the introduction is exactly the contribution-to-evaluation mismatch that commonly causes rejection.

F. Cross-Module Evaluation Protocol

The four models were trained and evaluated on different machines under different protocols. The figures in Tables IV–VI are therefore not strictly comparable across modules: they use the same metric definitions but different test-set sizes and different evaluation conditions. This is stated explicitly rather than presenting the tables as a unified benchmark.

VI. Discussion

A. What the Results Support

The behaviour and object detectors reach mean average precision at the 0.50 threshold above 0.90, which is adequate for a decision-support role in which every alert is reviewed by a human operator before action is taken. The swimmer detector's performance is lower but sufficient to produce occupancy estimates at the granularity that the maintenance schedule requires: the thresholds are expressed in tens of person-hours, so a small persistent bias in the count shifts a recommendation by a bounded amount rather than changing its direction.

B. What the Results Do Not Support

None of the modules should be described as replacing the function it assists. The behaviour detector misses approximately 12% of drowning instances per frame and does not replace lifeguard supervision. The water-quality classifier has not been shown to generalise beyond synthetic data and does not replace chemical testing. The maintenance schedule is derived from unvalidated thresholds and does not replace a facility maintenance policy. The system is an operator decision-support prototype.

C. On Data Quantity Versus Data Consistency

The swimmer detection comparison suggests that adding a second data source did not improve generalisation. This is consistent with the general observation that annotation consistency and domain match matter alongside dataset size, but the present experiment is not sufficient to isolate the cause [CITATION NEEDED].

D. On Explainability

The maintenance layer is rule-based by design. Every recommendation can be traced to a specific accumulated person-hour value crossing a specific named threshold, and the operator can be shown that arithmetic. Replacing this layer with a learned scheduler would likely improve nothing that matters at this scale while removing the property that makes the output auditable.

E. Comparison with Existing Work

[VERIFY] Once Section II is populated, add a short quantitative comparison against the closest reported system: state the comparison numerically and state what is not comparable (different datasets, different test protocols). Do not claim to outperform a system evaluated on a different dataset.

VII. Limitations and Threats to Validity

1) Perception: Swimmers overlap and become partially submerged; surface reflection and variable illumination affect detection; small and distant swimmers are missed. Models trained on one pool may not transfer to another.

2) Counting: The system measures per-frame occupancy, not unique entries and exits. Identity tracking is not implemented, so a person leaving and re-entering is counted as present throughout. Median smoothing introduces a bounded delay before genuine occupancy changes are reflected.

3) Evaluation: Test-set sizes are small or unstated. The four modules were evaluated under different protocols on different machines. No confidence intervals or significance tests are reported. At minimum, the test-set size should accompany each metric so the reader can judge the precision of the estimate.

4) Reproducibility: The committed swimmer training script does not reproduce the deployed checkpoints. Dataset provenance and licences for Component 1 are not documented. Random seeds are not recorded.

5) Deployment: Maintenance thresholds and the area-per-bather constant are unvalidated project defaults. The pool polygon requires per-installation calibration. Bather-load history is held in memory and does not survive a restart. Real-time throughput and latency are unmeasured. Privacy, security, failure recovery and operator override behaviour have not been evaluated.

6) Simulation: The system includes a simulation mode that generates synthetic counts when a model or camera is unavailable. This exists solely to exercise the interface and is never used as evidence of model performance. No figure in this paper derives from simulation mode.

VIII. Conclusion and Future Work

This paper presented an integrated swimming pool monitoring framework combining swimmer detection, behaviour recognition, foreign-object detection and water-chemistry classification behind a shared acquisition layer and a cross-module decision layer. The distinguishing element is the conversion of per-frame occupancy into accumulated bather load in person-hours, which drives a transparent threshold-based maintenance schedule and thereby carries a perception output through to an operational decision. The behaviour and object detectors achieved mean average precision at the 0.50 threshold of 0.907 and 0.903 respectively. The comparison between single-source and combined-source swimmer training data produced no measurable improvement, indicating that additional data volume alone did not improve generalisation in this setting.

Future work follows directly from the limitations. In order of priority: establish a single evaluation protocol with stated test-set sizes across all four modules; document dataset provenance and licences; reconcile the training script with the deployed checkpoints and record seeds, configurations and environment for every run; measure count mean absolute error, frame rate and end-to-end latency on the deployment hardware; evaluate the water-quality classifier against a threshold baseline and on out-of-distribution inputs; add multi-object tracking if unique-visitor counting is required; add persistent storage for bather-load history; and validate the maintenance thresholds against real facility maintenance logs and operator judgement.

Acknowledgment

The authors thank the Faculty of Computing, Sri Lanka Institute of Information Technology, for institutional support throughout this project.

[NOTE] The supervisors currently appear in the author block. If they remain there, they should not also be thanked here. Choose one.

References

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Las Vegas, NV, USA, 2016, pp. 779–788.

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[3] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” 2023, GitHub repository. [Online]. Available: https://github.com/ultralytics/ultralytics

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[5] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in Proc. Eur. Conf. Comput. Vis. (ECCV), Zurich, Switzerland, 2014, pp. 740–755.

[CITATION NEEDED] Only the five entries above are verified real publications. A conference paper of this scope would normally carry 20–30 references. The remaining entries must be built from papers the authors have located and read — not from memory and not from another paper's reference list without checking.

EDITORIAL NOTES — DELETE THIS PAGE BEFORE SUBMISSION

Status of this document

This is a complete structural draft in IEEE conference format. It is NOT submittable. Every bold bracketed note in the body marks a fact that must be supplied from your own records, or a citation that must be found and read. Nothing in this paper has been invented; where a value was not available, the gap was left visible rather than filled.

Blocking items — the paper cannot be submitted until these are done

1. Re-run the Component 1 evaluation per checkpoint on the same test split and repopulate Table IV from the saved raw output. Two contradictory metric sets currently exist and neither is supported by a results file.

2. Document Component 1 dataset sources, owners, URLs, licences and split sizes.

3. State test-set sizes for all four modules.

4. Build the reference list to 20–30 real, read sources. Five are currently verified.

5. Correct the v1-versus-v2 claim in the project README so it matches Table IV. The README currently asserts a dataset-heterogeneity finding that a 0.001 mAP difference cannot support.

6. Either answer RQ3 with timing measurements or remove RQ3 from Section I.

7. Run the water-quality threshold-baseline comparison, or narrow that module's claim.

8. Draw Fig. 1 (system architecture). The paper currently has no figures; a vision paper with none will read as incomplete.

9. Record training hardware, duration, library versions and random seeds for all four modules.

Formatting work still required in Word

This file approximates the IEEE conference layout: US Letter, 0.625 in side margins, two columns, Times New Roman, 10 pt body, 24 pt title, Roman-numeral level-1 headings in small caps, table titles above tables in Roman numerals, no colour fills. Download the official IEEE conference template from ieee.org/conferences/publishing/templates and paste this content in as unformatted text, then apply the template's own styles. Do not submit this file directly.

Equations (1)–(4) are rendered here as plain text. Retype them in Word's equation editor: IEEE rejects equations pasted as images, and Roman variables must be italicised while Greek symbols stay upright.

Six tables is a lot for a six-page paper. Consider merging Tables V and VI into one results table once the Component 1 numbers are confirmed.

Confirm from the ICAC 2026 call for papers

Page limit; submission deadline; whether review is double-blind (if so, the entire author block must be removed); and whether the venue specifies the A4 or US Letter template. This file is US Letter.

Author block correction already applied

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